

Smart Traffic Lights using Local AI

Student Handout - AI-Assisted Research & L^AT_EX Workshop

Abstract—Smart traffic-light systems are being studied to reduce waiting times and adapt rapidly to dynamic road conditions. This paper provides a literature review comparing traditional fixed-timer traffic signals, cloud-based AI systems, and local edge-based smart cameras. We outline a standard four-step system architecture, evaluate system performance metrics, and highlight key research gaps regarding emergency vehicle prioritization.

Index Terms—Artificial Intelligence, Edge Computing, IEEE Style, Smart Traffic Lights, Traffic Signal Control.

I. INTRODUCTION

Traditional traffic lights usually follow fixed timers. While simple and inexpensive, they cannot easily react when a single road experiences sudden congestion. Modern adaptive systems leverage cameras and artificial intelligence (AI) to estimate vehicle density and dynamically adjust signal timing. Instead of relying strictly on clock cycles, decisions are informed by real-time roadway metrics. Recent reviews confirm a growing interest in AI-based traffic signal control and its potential benefits under changing conditions [1].

II. LITERATURE REVIEW

A. Cloud vs. Local Processing

Cloud-based AI systems transmit camera feed data to a central server for processing. Although this offers high computing capacity, it introduces dependency on stable internet connectivity and increases communication latency.

Conversely, local smart-camera (edge computing) systems execute processing near the roadway. This minimizes bandwidth consumption and supports faster decision-making. Small local computers analyze localized camera feeds to issue immediate instructions to the signal controller. Machine-vision research highlights the efficiency of camera-based adaptive signal modes [2].

B. Critical Evaluation of Microcontroller Capabilities

Literature indicates extreme claims regarding low-cost microcontrollers. For example, Smith et al. [3] presented a claim where a basic Arduino Uno processed 10,000 live HD video streams simultaneously with zero delay. When evaluating such literature, it is essential to distinguish between actual hardware computational limits (such as 8-bit, 16 MHz hardware constraint boundaries) and theoretical or simulated claims.

III. SYSTEM ARCHITECTURE

The proposed system architecture is structured into four sequential steps:

- 1) **Street Cameras:** Capture live traffic conditions on-site.
- 2) **Smart Pole Computers:** Perform localized video processing and issue rapid traffic management decisions.

- 3) **City Cloud Server:** Receives selected summarized data for long-term storage and system-wide analytics.
- 4) **Traffic Lights:** Receive actuation decisions to adjust signal indicators.

The data and decision flow is summarized as follows:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

IV. PERFORMANCE COMPARISON

Table I summarizes the three primary traffic light system paradigms based on response time, bandwidth demand, and installation costs.

TABLE I
COMPARISON OF TRAFFIC SYSTEM ARCHITECTURES

System Type	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30.0 s	Zero	\$200
Cloud AI Systems	2.0 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

V. RESEARCH GAP

Existing traffic signal control solutions function effectively under regular, predictable daily commuting patterns. However, a significant research gap persists in emergency scenarios: many existing systems lack integrated protocols to automatically clear paths for approaching emergency vehicles, such as speeding ambulances. There is a distinct need for smart traffic architectures capable of detecting priority vehicles and dynamically coordinating surrounding intersections to route them safely.

VI. METHODOLOGY

To address the limitations in current adaptive traffic management and resolve the identified research gaps, a multi-phase methodology combining computer vision and edge computing is implemented:

A. Video Feed Acquisition and Preprocessing

High-definition optical sensors mounted on local smart poles capture continuous video streams from each intersection approach. Frame extraction occurs locally at a rate of 15 frames per second (fps) to balance real-time tracking with thermal and processing constraints on edge devices.

B. Edge Inference and Object Detection

A lightweight deep neural network model (e.g., YOLOv8-Nano optimized via TensorRT) is deployed directly on the localized smart pole single-board computers (SBCs). The neural network performs two parallel tasks:

- **Vehicle Density Calculation:** Detects standard vehicles (cars, buses, motorcycles) and calculates bounding box concentrations to compute lane-specific congestion indexes in real time.
- **Priority Vehicle Classification:** Scans frames for optical markers and emergency features (such as ambulance flashing lights and vehicle geometry) to trigger an immediate interrupt signal.

C. Dynamic Signal Timing and Emergency Priority Logic

The edge controller evaluates the vehicle density index every 500 ms using a local decision algorithm:

- 1) **Standard Mode:** Allocates green light duration proportionally based on comparative lane density metrics while maintaining minimum safety thresholds for pedestrian crossings.
- 2) **Preemption Mode:** Upon local detection of an emergency vehicle, the algorithm overrides normal cycles, rapidly clears conflicting signals, and forces an immediate green signal along the emergency vehicle's vector. Inter-intersection coordination messages are sent via lightweight MQTT protocol to prepare downstream intersections.

VII. CONCLUSION

This paper reviewed traditional, cloud-assisted, and edge-based smart traffic light systems, highlighting the superior response times (0.5 s) and reduced bandwidth demands of localized processing architectures. We proposed an integrated methodology incorporating computer vision on local smart pole edge devices to dynamically regulate traffic density and preemptively clear signals for approaching emergency vehicles. Future research will focus on deploying this local AI framework across multi-intersection testbeds to quantify real-world reductions in emergency response times and overall urban traffic latency.

REFERENCES

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